

MICROPROCESSOR AND INTERFACING

Microprocessor Architecture & Peripheral Interfacing

8086 Instruction Set, Memory Segmentation, Interrupt Handling, Bus Timings & Peripheral Chips (8255, 8259, 8253)

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CHAPTER 1

8086 Internal Architecture, Registers & Memory Segmentation

1. Theoretical Foundations & Scope

The 8086 16-bit microprocessor represents a landmark architecture in modern computing. This chapter dissects the Bus Interface Unit (BIU), Execution Unit (EU), 20-bit physical address calculation, and 64KB segmented memory organization.

2. Core Conceptual Framework

- **BIU and EU Pipelining:** Instruction prefetch queue enabling overlapping fetch and execution cycles.
- **Memory Segmentation:** Logical segment:offset address generation multiplying segment register values by 16 to produce 20-bit physical addresses.
- **Register File Organization:** General-purpose (AX, BX, CX, DX), pointer/index (SP, BP, SI, DI), and flag registers.

Key Practical Insights & Examination Takeaways:

- ' Calculate 20-bit physical addresses from CS:IP, DS:BX, and SS:SP pairs.
- ' Explain the operational advantages of instruction pipelining.
- ' Analyze status and control flags including Zero, Carry, Sign, and Overflow flags.

CHAPTER 2

Assembly Language Programming & Addressing Modes

1. Theoretical Foundations & Scope

Programming at the assembly level gives precise control over hardware resources. We study the eight 8086 addressing modes, data transfer instructions, arithmetic/logic operations, branch controls, and string primitives.

2. Core Conceptual Framework

- **Addressing Modes:** Immediate, register, direct, register indirect, based, indexed, based-indexed, and relative modes.
- **Stack Operations & Subroutines:** CALL and RET mechanics, parameter passing via stack frames, and interrupt stack preservation.
- **String Instructions:** MOVSB, CMPSB, SCASB with REP prefixes leveraging SI, DI, and CX counter registers.

Key Practical Insights & Examination Takeaways:

- ' Write efficient assembly routines for array manipulation and arithmetic operations.
- ' Construct modular subroutines with balanced push/pop stack frames.
- ' Debug low-level register states using simulator step debugging.

CHAPTER 3

Peripheral Interfacing: 8255 PPI, 8259 PIC & 8254 Timer

1. Theoretical Foundations & Scope

Processors interact with the physical world through peripheral controllers. This chapter explores memory-mapped vs I/O-mapped I/O, programmable peripheral interface (8255), programmable interrupt controller (8259), and counter/timer (8254).

2. Core Conceptual Framework

- **8255 PPI Modes:** Mode 0 (Basic I/O), Mode 1 (Strobed I/O with handshaking), and Mode 2 (Bi-directional bus).
- **8259 Interrupt Controller:** Interrupt vector tables, priority resolver, and interrupt service routine (ISR) vectoring.
- **DMA & ADC/DAC Interfacing:** Direct Memory Access bypassing processor overhead, and analog-to-digital converter interfacing.

Key Practical Insights & Examination Takeaways:

- ' Configure 8255 control words for custom input/output port assignments.
- ' Explain hardware interrupt priority resolution and vector generation.
- ' Interface stepper motors, seven-segment displays, and keypads to microprocessors.

Academic References & Further Study

[1] Quality Education Academic Standards & Pedagogical Guidelines, 2026 Edition.

[2] International Computer Science & Engineering Curriculum Framework.

[3] Educational Technology & Assessment Methodology Research Series.

[4] Open Access Educational Resource Repository - Quality Education Platform.

